ಪಿ.ಡಿ.ಎ. ತಾಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ

ಎಐಸಿಟಿ ನವದೆಹಲಿ ಇಂದ ಮಾನ್ಯತೆ ಪಡೆದ – ವಿಟಿಯು ಬೆಳಗಾವಿ ಸಂಯೋಜಿತ ಸ್ವಾಯತ್ತ ಸಂಸ್ಥೆ

PDA COLLEGE OF ENGINEERING

Autonomous Institute Affiliated to VTU Belagavi - Approved by AICTE New Delhi

KALABURAGI
ESTD: 1958

Major Project-1

AI powered Intelligent diet recommendation and calorie tracking system

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Abstract

Maintaining a healthy and balanced diet is crucial for overall well-being, but many individuals face difficulties in tracking their calorie intake and planning meals efficiently. Manual calorie counting is often inaccurate and time-consuming, while generic diet plans fail to meet specific nutritional needs. This project proposes a Diet Recommendation System with Calorie Tracking and Smart Meal Planning that provides personalized meal plans, automated calorie tracking, and AI-driven diet recommendations. By integrating nutrition databases, machine learning algorithms, and web technologies, the system ensures a tailored dietary experience based on user preferences, health conditions, and fitness goals. The application is designed to be user-friendly, scalable, and adaptable to different dietary patterns, helping individuals make informed nutritional choices and maintain a healthier lifestyle effortlessly

Contents

1.Introduction

Maintaining a well-balanced diet is essential for overall health, but many individuals struggle with tracking their daily calorie intake and planning meals effectively. Traditional diet planning methods often lack personalization, making them ineffective for users with specific health goals, dietary restrictions, or lifestyle preferences. Moreover, manual calorie tracking can be time- consuming and prone to errors.

To address these challenges, this project aims to develop an intelligent diet recommendation system that suggests personalized meal plans, tracks calorie intake, and offers smart meal planning based on user preferences and health requirements. By leveraging web technologies and AI-driven algorithms, the system will simplify nutrition management, helping users achieve their dietary goals more effectively.

The Need for Smart Diet Systems

Rising Obesity Rates

Obesity is a global epidemic. It affects over 650 million adults.

Diet-Related Diseases

Diseases like heart disease and diabetes cost ₹20.88 trillion annually.

Traditional Dieting Limits

Traditional diets often lack personalization and adherence.

The Diet Recommendation System offers a smarter approach to dieting. It addresses these critical limitations.

2. Objectives

- **Personalized Nutrition Guidance:** Develop an intelligent system that generates customized meal plans based on the user's health profile, dietary restrictions, and fitness goals.
- **Automated Calorie Tracking:** Integrate a nutrition database and food recognition APIs to automate calorie calculations, reducing manual effort.
- **Smart Meal Planning:** Implement an AI-driven meal planner that suggests optimized meal combinations based on available ingredients and daily nutritional requirements.
- **Health Monitoring & Diet Optimization:** Continuously analyze user data to provide real-time diet suggestions and adaptive meal plans that evolve with progress.
- **Food Recommendation System:** Suggest healthier alternatives when users log high-calorie or unbalanced meals.
- **User Progress Tracking:** Enable users to monitor their dietary habits through an interactive dashboard that displays nutritional trends and caloric balance.

3.Problem Statement

The Growing Challenge of Personalized Nutrition

In today's busy and fast-moving world, many people struggle to maintain a healthy diet. Busy work schedules, lack of nutritional knowledge, and limited time for meal planning often lead to unhealthy eating habits. This problem is becoming more serious as we see a growing number of lifestyle-related health issues such as obesity, type 2 diabetes, and heart diseases. These conditions are directly linked to poor nutrition and can significantly affect a person's quality of life.

Traditional diet plans often fail to meet individual needs. They are usually one-size-fits-all and don't consider important personal factors like age, gender, daily activity level, food allergies, or cultural and regional eating habits. For example, a working adult in a metropolitan city may have very different dietary requirements compared to a student living in a small town, but most diet plans don’t reflect these differences.

Moreover, keeping track of daily calorie intake, nutrients, and meal balance manually can be overwhelming and time-consuming. It’s easy to make mistakes or lose motivation, which makes it hard for people to stay consistent.

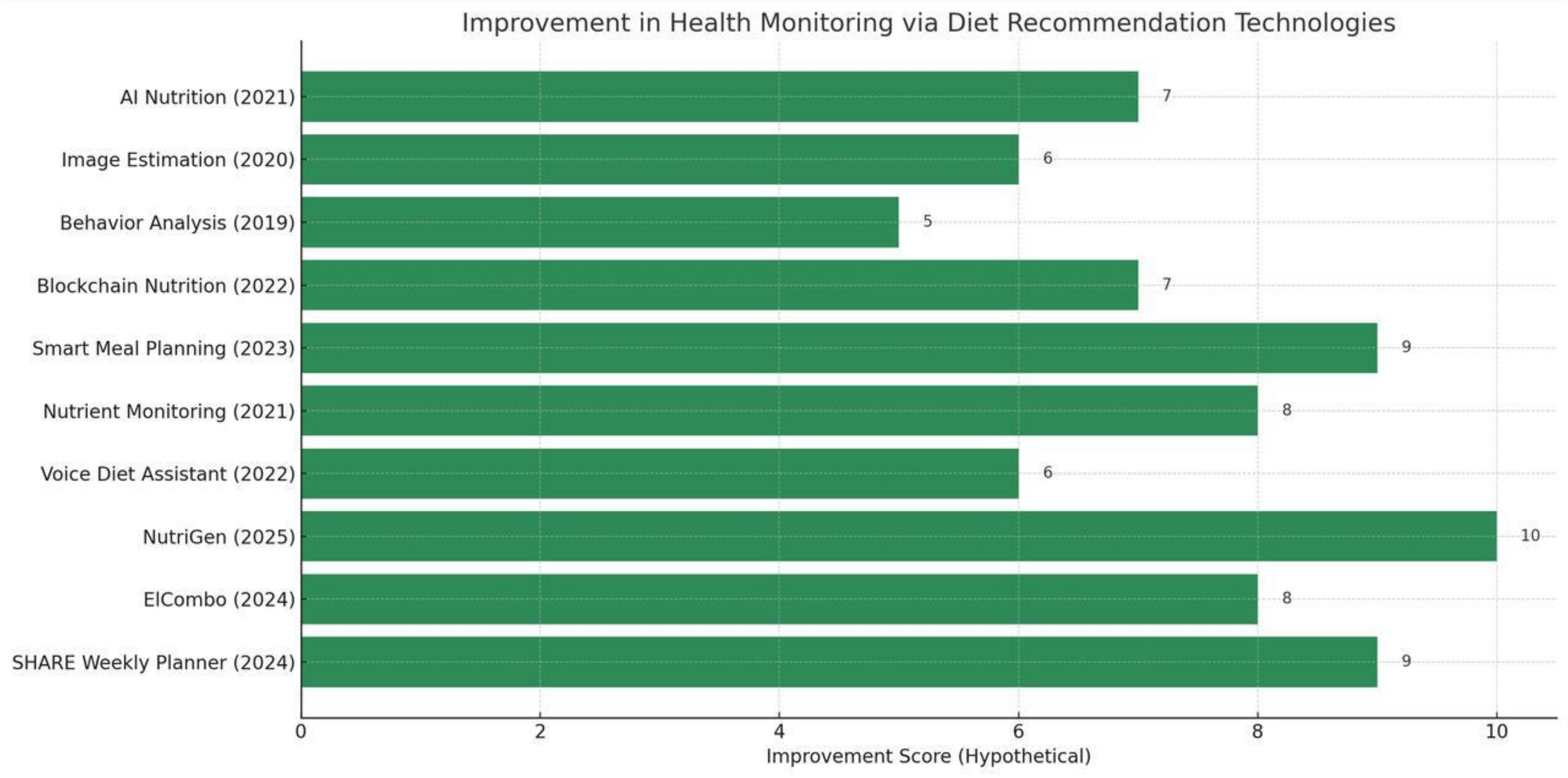
Although artificial intelligence (AI) has made great progress in other fields, its integration into personal nutrition and fitness planning is still limited. There is a growing need for smart systems that can provide real-time, personalized advice—adapting to changing goals, preferences, and lifestyles—to help individuals make healthier choices without added stress or confusion.

4. Literature Survey

Comparison of Existing Research Studies on Diet Recommendation Systems

No.	Study	Technology Used	Research Focus	Applications	Limitations
1	AI-Based Personalized Nutrition System (2021)	Machine Learning (ML)	Personalized meal recommendations based on past food habits	AI-driven diet coaching apps	Requires large training data, slow adaptation
2	Calorie Estimation Using Image Recognition (2020)	CNN (Computer Vision)	Estimating calorie intake using food image analysis	Mobile calorie tracking apps	Accuracy depends on image quality and food database
3	Dietary Tracking and Behavioral Analysis (2019)	Data Analytics	Long-term tracking of eating habits	Habit-forming diet apps	Requires continuous manual input
4	Nutrition Recommendation System Using Blockchain (2022)	Blockchain	Secure diet tracking with tamper-proof records	Medical diet tracking systems	High computational cost
5	AI-Enhanced Smart Meal Planning (2023)	AI-based meal planning optimization	Adjusting meal plans based on food availability and calories	Grocery and meal kit delivery apps	Needs constant updates to food databases

No.	Study	Technology Used	Research Focus	Applications	Limitations
6	Smart Nutrient Intake Monitoring (2021)	IoT & Wearable Devices	Tracking nutrients through smart sensors	Integration with fitness trackers and smartwatches	High cost of wearable devices
7	Voice-Activated Diet Assistant (2022)	NLP (Natural Language Processing)	Voice-based diet logging and recommendations	AI-powered virtual assistants like Alexa & Google Assistant	Limited voice recognition accuracy for complex meal descriptions
8	NutriGen: Personalized Meal Plan Generator Leveraging Large Language Models (2025)	Large Language Models (LLMs)	Generating personalized meal plans aligning with user-defined dietary preferences and constraints	AI-driven personalized nutrition planning	Requires extensive user data and computational resources
9	ElCombo: Personalized Meal Recommendation System for Chinese Older Adults (2024)	Knowledge Graphs, Personalized Algorithms	Providing meal recommendations considering disease-related nutritional constraints and personal dietary preferences	Enhancing dietary quality among older adults	Limited to specific demographic and cultural dietary data
10	SHARE: Healthy Personalized Recipe Recommendations for Weekly Meal Planning (2024)	Collaborative Filtering, Content-Based Filtering	Offering comprehensive meal plans considering individual dietary tastes, health histories, and dietary restrictions	Personalized weekly meal planning applications	May require continuous user feedback for optimal performance


Improvement in Health Monitoring via Diet Recommendation Technologies
AI Nutrition (2021)
7
Image Estimation (2020)
6
Behavior Analysis (2019)
5
Blockchain Nutrition (2022)
7
Smart Meal Planning (2023)
9
Nutrient Monitoring (2021)
8
Voice Diet Assistant (2022)
6
NutriGen (2025)
10
ElCombo (2024)
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SHARE Weekly Planner (2024)
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Improvement Score (Hypothetical)

Fig.1

5. Requirement Analysis

Functional Requirements:

- User Registration and Profile Management (age, weight, dietary preferences).
- Personalized Meal Planning (customized diet suggestions).
- Calorie Tracking System (automated intake monitoring).
- Smart Meal Planner (meal suggestions based on available ingredients).

Non-Functional Requirements:

- **Scalability:** Should handle multiple users efficiently.
- **Security:** Protect user data using secure authentication.
- **Usability:** Ensure an intuitive UI with responsive design.

Technology Stack:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python (Flask/Django) or Node.js
- **Database:** MySQL or Firebase
- **API Integration:** Nutrition APIs (for food data and calorie counts)

6. Proposed Design / Methodology

The Diet Recommendation System will be developed using a modular approach, ensuring flexibility, scalability, and adaptability. The system consists of the following core components:

1.User Profile Setup

- Users enter basic details (age, weight, height, activity level, dietary preferences, and health goals).
- The system calculates Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE) to determine personalized calorie requirements.

2.AI-Powered Diet Recommendation Engine

- Uses machine learning models to analyze user input and suggest customized meal plans based on macronutrient balance (protein, carbs, fats).
- Integrates with nutrition APIs to fetch calorie and nutrient values of foods.

3.Smart Meal Planning System

- Suggests meals based on available ingredients in the user's kitchen (via user input or grocery list integration).
- Users can modify meal plans dynamically based on preferences.

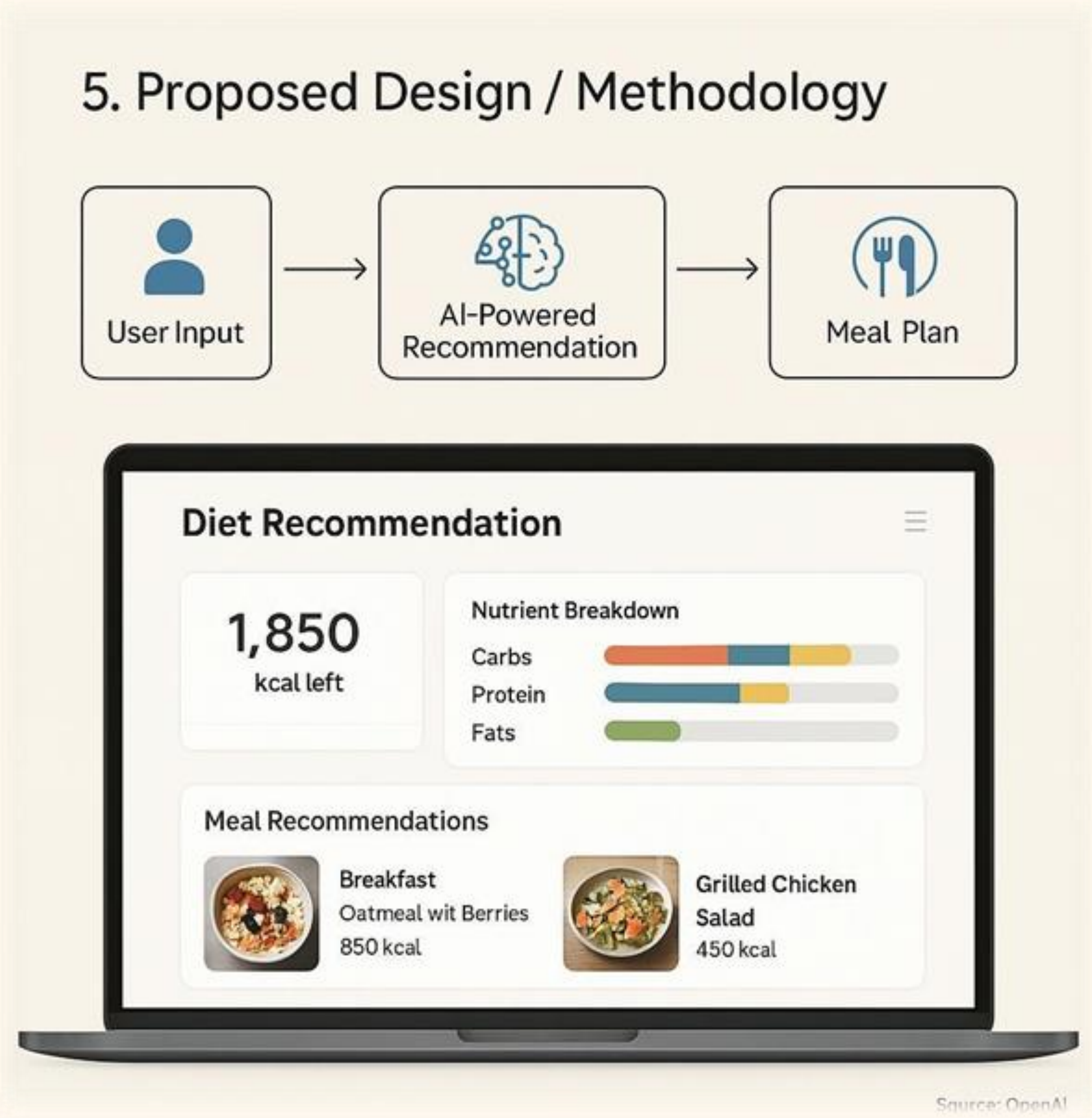


Fig.2

4.Automated Calorie Tracking

- Integrates food recognition APIs (e.g., FoodData Central, Edamam) to estimate calories.
- Allows manual food logging for items not found in the database

5.User Dashboard & Health Insights

- Displays daily calorie intake, nutrient breakdown, and meal plan adherence.
- Provides dietary trend analysis and personalized health tips.

6.Adaptive Learning & Feedback System

- Uses AI-based feedback loops to refine meal recommendations based on user behavior.
- Suggests alternative meals if a user skips or modifies their diet frequently.

7.Cross-Platform Integration

- Web-based UI using HTML, CSS, JavaScript.
- Backend with Python (Flask/Django) or Node.js.
- Integration with mobile apps and wearable devices for seamless calorie tracking.

Algorithm: Smart Diet+ Recommendation System

Step 1: Start

Step 2: Collect user input from the web form:→ Age, Gender, Occupation, Diet Preference, City, Allergies

Step 3: Send the input data to the Flask backend via POST request (/recommend route)

Step 4: In the backend:
a.Validate input data
b.Normalize data (e.g., lowercase diet preferences, trim whitespace)

Step 5: Analyze user profile using logic or a lightweight AI model: Check age group for calorie needs Match diet preference with available meal plans Consider city for local food availability (optional)Remove any options that include allergens

Step 6: Select a suitable meal recommendation from the dataset or logic rules

Step 7: Select a matching exercise plan based on Age Occupation (e.g., sedentary vs active)User goals (optional: e.g., weight loss, maintenance)

Step 8: Prepare final response (Smart Meal + Smart Exercise Plan)

Step 9: Return the result to the frontend and display it to the user

Step 10: End

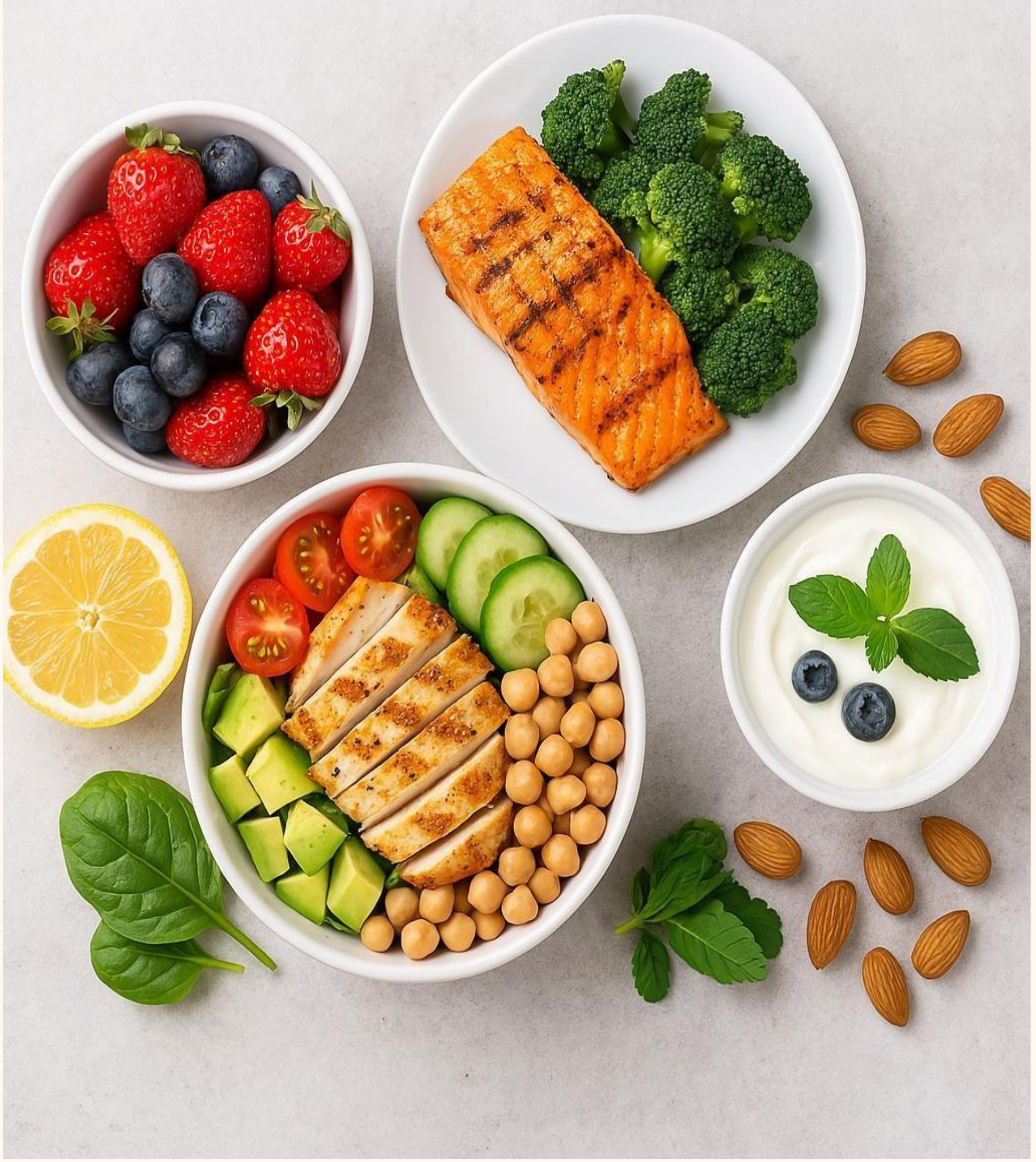
Fig.3

7. Conclusion

The Diet Recommendation System presents a smart, personalized approach to meal planning and calorie tracking, integrating machine learning, AI-driven recommendation models, and automated tracking mechanisms to simplify diet management.

This system goes beyond conventional meal planning by incorporating real-time feedback loops, which enable continuous improvements based on user behavior. The nutrition database integration ensures that dietary recommendations are well-informed and aligned with standard health guidelines. Moreover, adaptive meal planning dynamically adjusts meal suggestions based on the user's dietary preferences, health goals, and past consumption patterns.

By leveraging these advanced features, the system becomes an efficient and intuitive tool for individuals striving to maintain a healthy lifestyle, offering data-driven, personalized, and adaptable meal planning for optimal nutrition and well- being.



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